

Project Introduction

Modern electrical distribution networks are experiencing significant transformations due to the integration of distributed energy resources such as photovoltaic (PV) systems. These components introduce variability and uncertainty in both generation and demand, affecting voltage stability and power losses throughout the network. Voltage magnitudes in distribution systems are sensitive to the spatial distribution of Distributed energy resources and to the placements and sizes of photovoltaic cells. In this project, we install PVs at random locations on a 33-node distribution network and vary their outputs to reflect realistic operating conditions. The resulting voltage magnitudes at each node are recorded for each scenario. The aim is to analyze how different DER configurations influence the network's voltage profile and identify reconfiguration strategies that can improve voltage regulation and Minimize active power losses. To achieve this, you will develop surrogate models to approximate the network behavior, evaluate their performance using test scenarios, and apply explainable AI (XAI) techniques to interpret, the most influential factors affecting voltage deviations across the network.

Dataset structure and preprocessing

VerilerPV.mat file convert to input.csv and output.csv files for ease of use

Input shape: (2000, 12)

Output shape: (2000, 33)

input features: [Tap1, Tap2, PV1loc, PV2loc, PV3loc, PV4loc, PV5loc, PV1size, PV2size, PV3size, PV4size, PV5size]

Output features : [|V1|, |V2|, ..., |V33|]

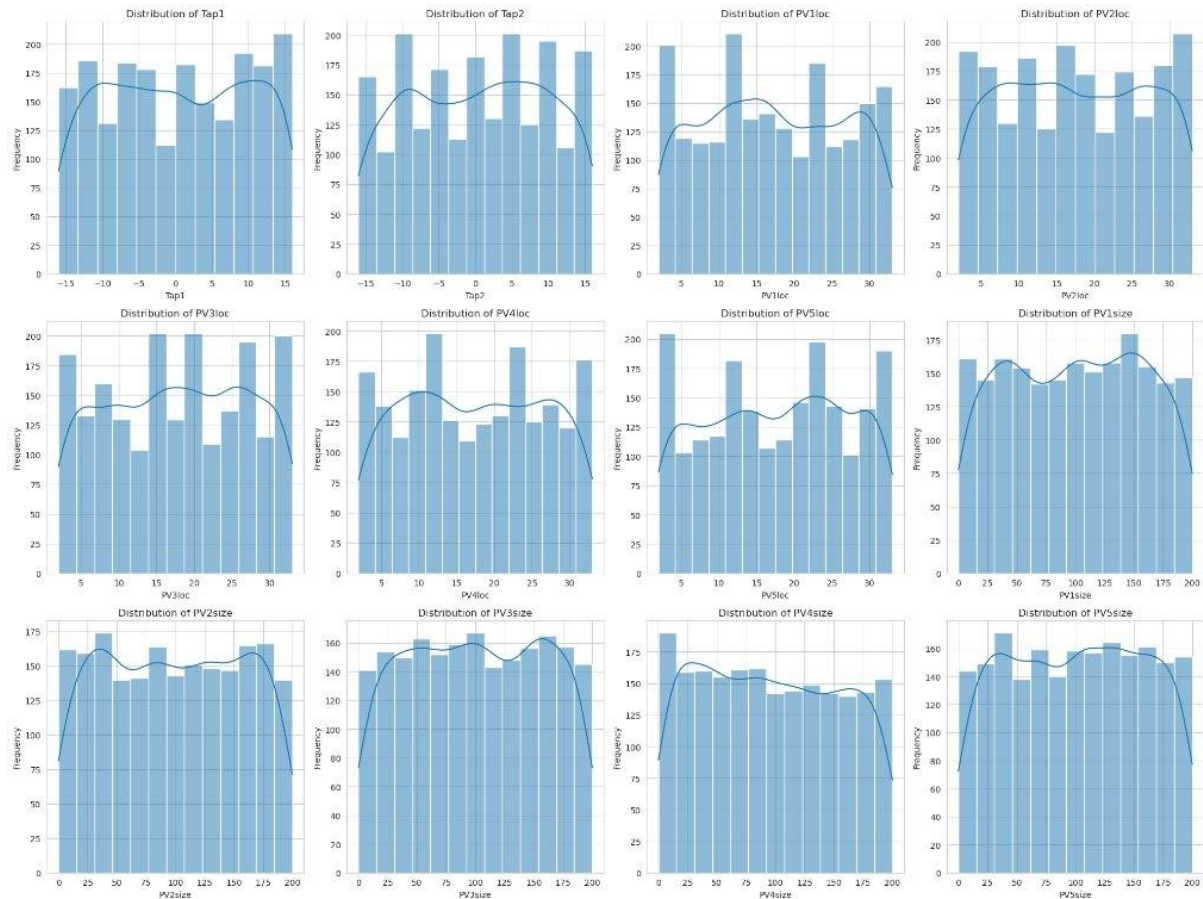
Information about Input Dataset

Tap1, Tap2 are the tap settings of two on-load tap changers

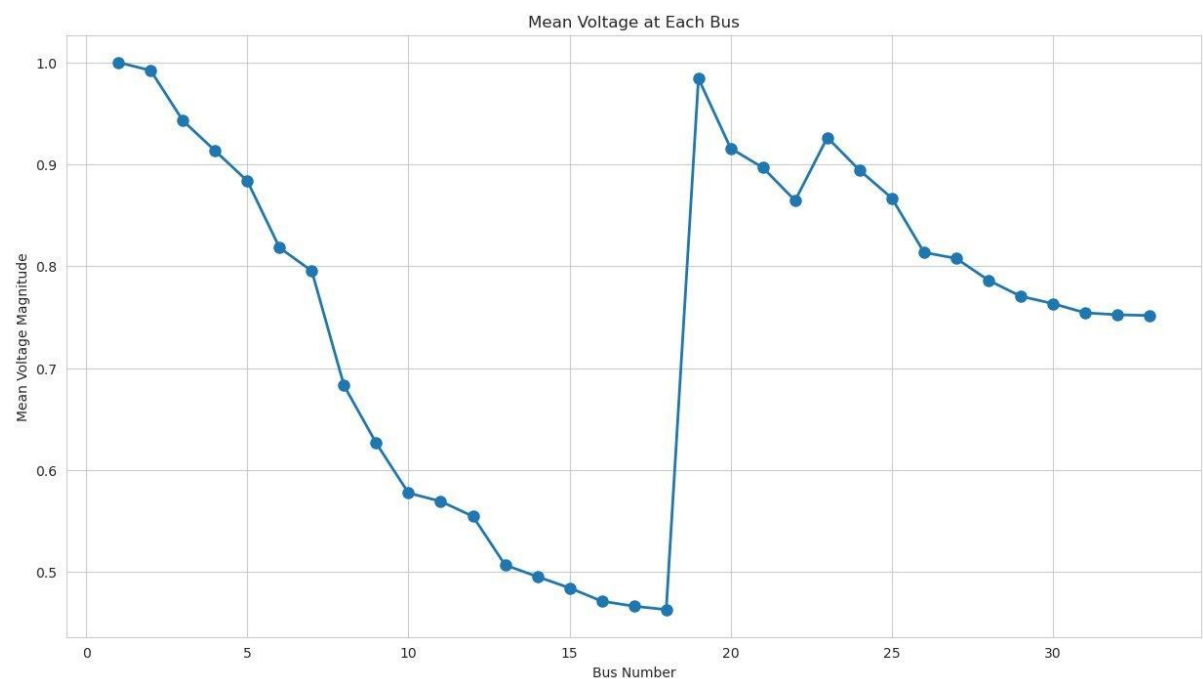
PV1loc – PV5loc are the bus IDs where photovoltaic (PV) units are placed,

PV1size – PV5size are the active power injection levels of each PV unit (e.g., in kW or normalized units).

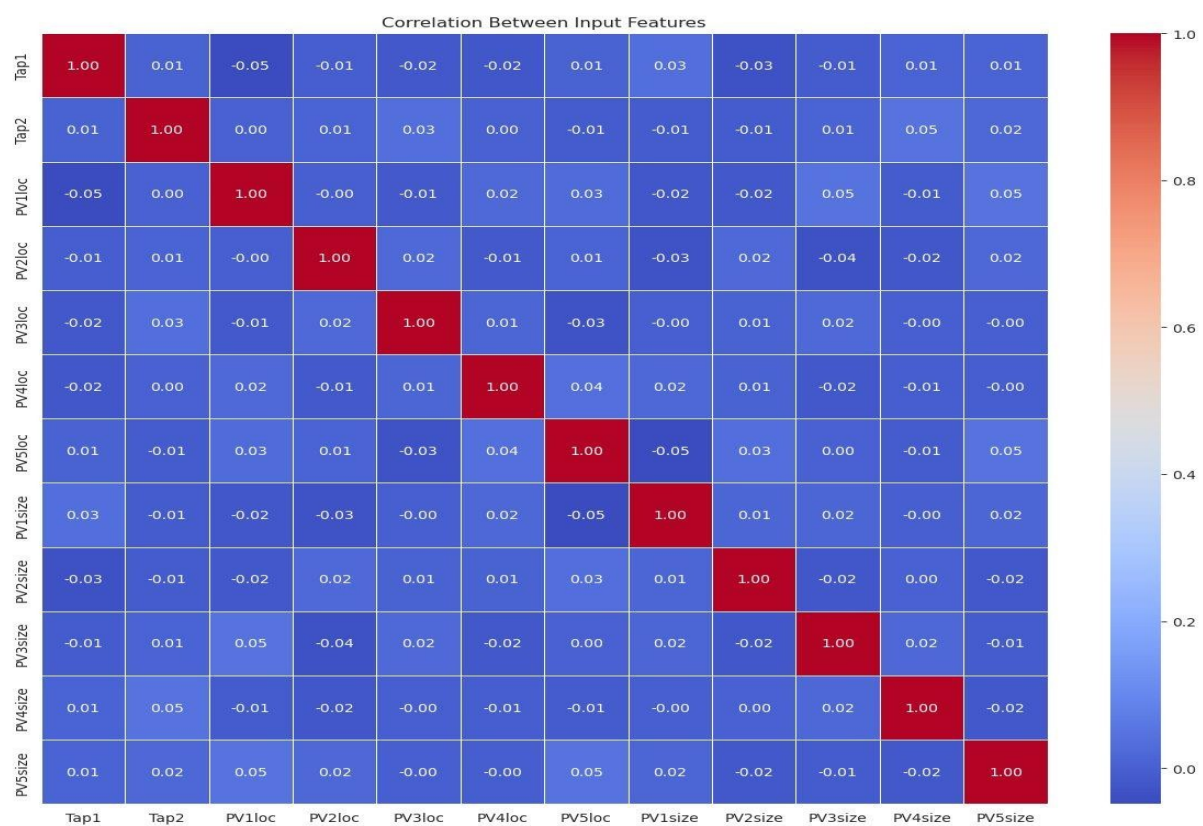
Information about Output Dataset, voltage magnitudes at all 33 buses of a power distribution network. Implemented Data manipulation, preprocessing and visualization. Basic statistics for input and output data (e.g. Count, Mean, Standard Deviation) visualization to distribution of input features. Distribution of input features look like uniform distribution



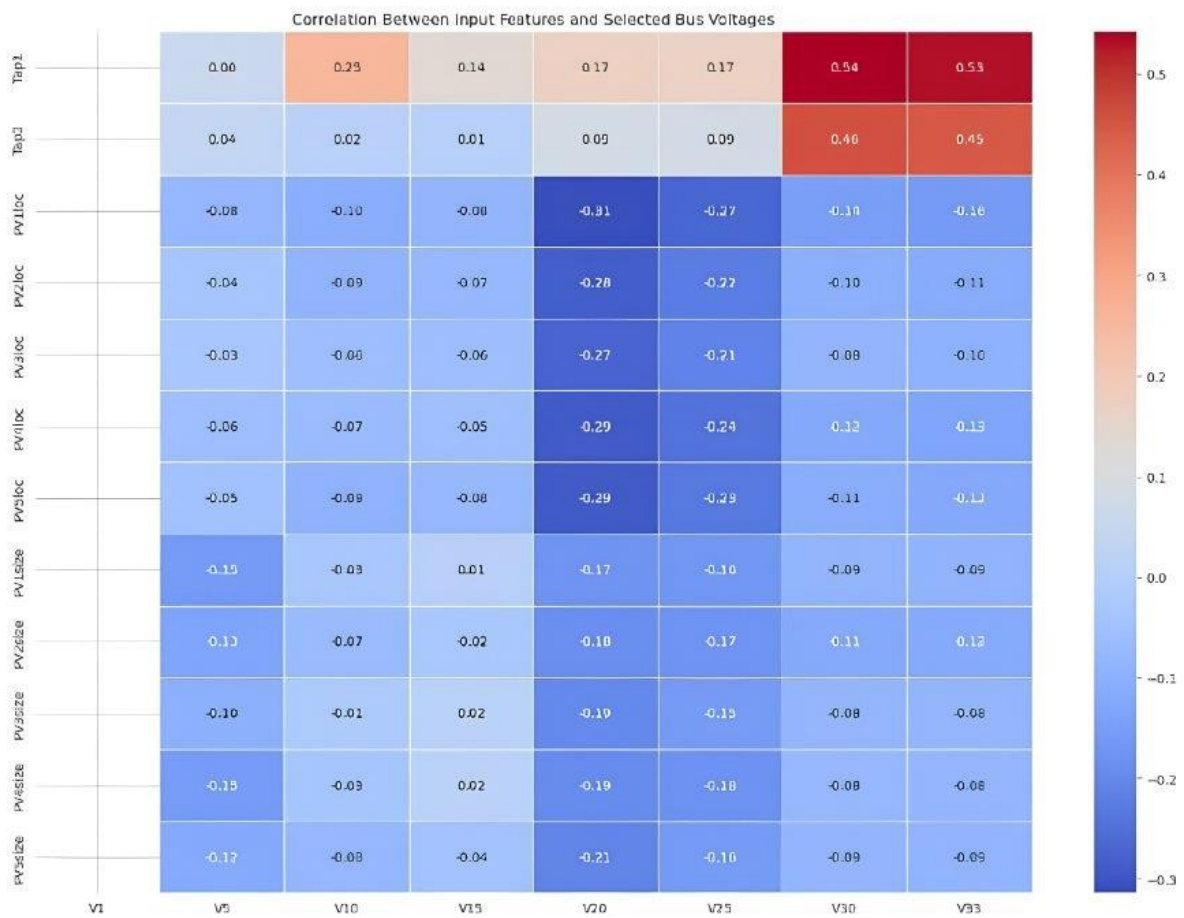
Distribution of output features



Initial decrease from Bus 1 to Bus 18, there's a monotonic decline in voltage magnitude, reaching a low of around 0.46 V. Sudden Spike at Bus 19, the voltage jumps back up near 0.98 V. This could indicate, generator or voltage regulator at Bus 19. A switching operation or reconfiguration. Gradual decline again, from Bus 19 to Bus 33, the voltage again gradually decreases and flattens out around 0.75–0.76 V This is not a classic statistical distribution (like Gaussian, Poisson, etc.), but we can describe its shape in terms of signal characteristics. It resembles a piecewise signal. First segment, almost exponential or quadratic decay. Jump/spike, At one specific index (Bus 19). Second segment, a damped oscillation or slightly noisy decreasing trend. Heat map of Correlation between input features is very smooth.



There is not any correlation between input features. But heat map of correlation between input features and output features give us many information about features' connections.



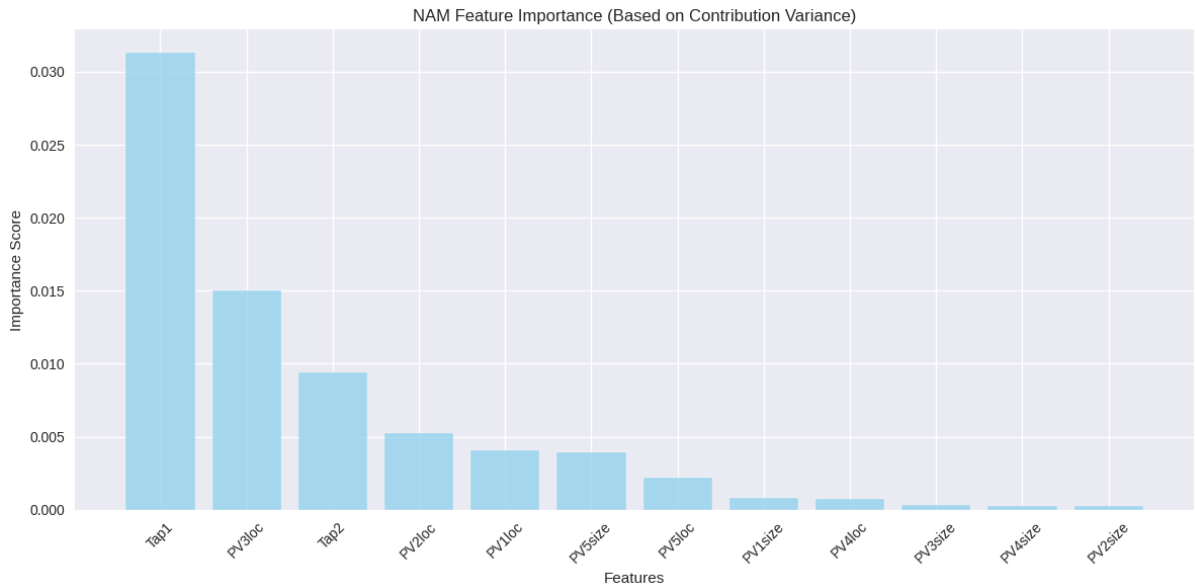
This connections tell us that how different DER configurations influence the network's voltage profile and identify reconfiguration strategies. Importantly, tap settings of two on load tap changers positively correlated with Bus Voltages. Then, Photovoltaic (PV) units' location negatively correlated with output Bus voltages. Remain features does not have any important correlate value. normalized input features using StandartScaler for clear and better result.

NAM architecture and training process

NAM architecture has 39,564 (154.55 KB) total parameters. Accordingly, each of FeatureNN has equally 3,297 parameters NAM has 12 feature-specific subnetworks for one per input feature. Output layer predict a 33-dimensional vector as multi-output regression for output 33

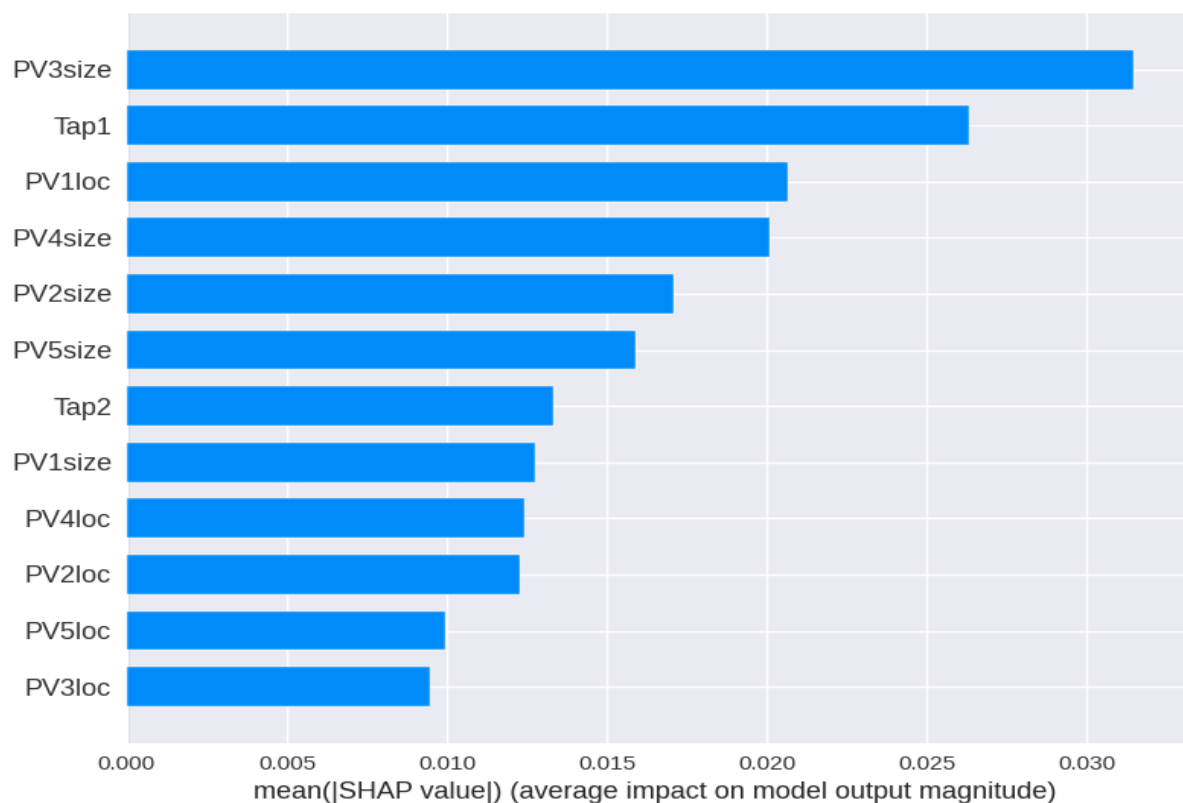
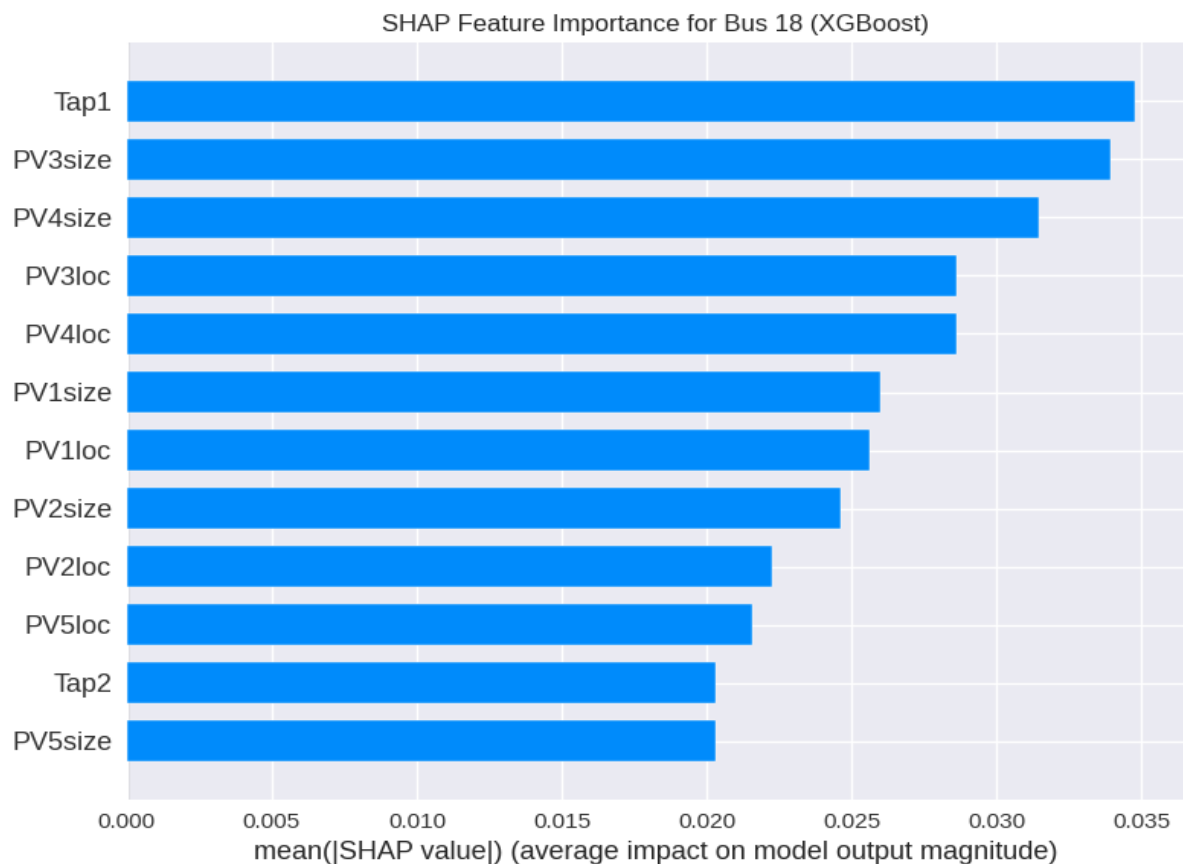
Individual Feature Network Architectures:			Model: "nam"		
Tap1 Network: Model: "Tap1_network"			Layer (type)	Output Shape	Param #
Layer (type)	Output Shape	Param #	FeatureNN_0 (FeatureNN)	?	3,297
Tap1_input (InputLayer)	[(None, 1)]	0	FeatureNN_1 (FeatureNN)	?	3,297
dense (Dense)	(None, 64)	128	FeatureNN_2 (FeatureNN)	?	3,297
dropout (Dropout)	(None, 64)	0	FeatureNN_3 (FeatureNN)	?	3,297
dense_1 (Dense)	(None, 32)	2080	FeatureNN_4 (FeatureNN)	?	3,297
dropout_1 (Dropout)	(None, 32)	0	FeatureNN_5 (FeatureNN)	?	3,297
dense_2 (Dense)	(None, 33)	1089	FeatureNN_6 (FeatureNN)	?	3,297
=====			FeatureNN_7 (FeatureNN)	?	3,297
Total params: 3,297			FeatureNN_8 (FeatureNN)	?	3,297
Trainable params: 3,297			FeatureNN_9 (FeatureNN)	?	3,297
Non-trainable params: 0			FeatureNN_10 (FeatureNN)	?	3,297
			FeatureNN_11 (FeatureNN)	?	3,297
			Total params: 39,564 (154.55 KB)		
			Trainable params: 39,564 (154.55 KB)		
			Non-trainable params: 0 (0.00 B)		

NAM model trained with 50 epoch. It is enough for Interpretation our problem. Tap1, PV3loc and PV3size features selected as representative input features. Bus Voltages 6, 18, 29 that representative output features selected because of particularly those values are best representative chosen buses at different locations. beginning, middle, end samples selected for better demonstration.

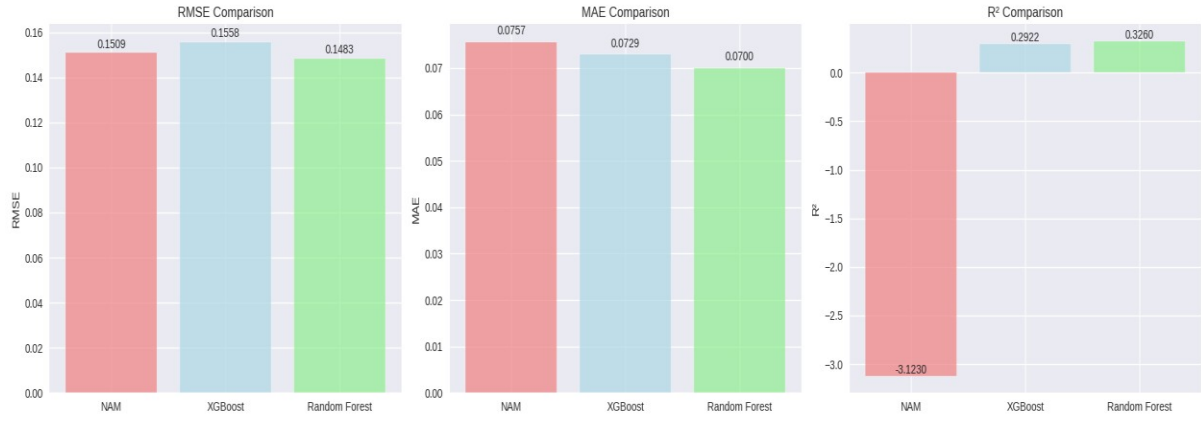


SHAP explanations for baselines

RandomForest and XGBoost regressors trained for implementing SHAP for explanation. MultiOutputRegressor wrapped to handle 33 dimensional output. dependence plots generated as SHAP-Based Visualizations. Random forest and XGBoost summary plots created for Global feature importance visualization. Summary plots include that contribution of selected representative input features on selected particular bus voltages. SHAP dependence plot created for input variable versus SHAP values for selected output.



Comparative tables and interpretation analysis



Conclusion: interpretability - performance trade-off

XGBoost provide more actionable insights for PV task. Because, It has best RMSE value. However, XGBoost can be problematic for different huge complex datasets and complex cases because of High CPU intensive. NAMs may be solution for this cases due to optimized with GPU/TPU. SHAP was able to recover similar patterns to the NAM plots but SHAP act as too different in particular data points. Especially for sudden breaking data points. There are not any discrepancies between the importance values and direction of effect that my observations.

Reference

Agarwal, R. ve diğerleri. N18 SHAP – Interpretable Machine Learningeural Additive Models: Interpretable Machine Learning with Neural Nets. NeurIPS 2021.

Scikit-Learn Kullanım Kılavuzu. MultiOutputRegressor – Çok Çıktılı Regresyon. (2025)

ApX Machine Learning. SHAP Summary & Dependence Plots – Model Interpretability Course. (2025)

SHAP – Interpretable Machine Learning